

Akshat Kankani

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EDUCATION

Manipal Institute of Technology

B.Tech in Computer Science (Artificial Intelligence) — CGPA: 8.11

Bengaluru, KA

July 2023 – May 2027

PROJECTS

AlphaLab | *Python, FastAPI, PostgreSQL, DuckDB, Redis, Celery, React*

Live Demo | GitHub

- Built a quantitative research platform for testing predictive financial factors against noise and imputations using reusable data pipelines and reproducible experiment workflows.
- Designed a custom DSL (lexer, parser, AST) that compiles factor formulas into vectorized computations, reducing backtest runtime vs. naive implementation.
- Built a bias-aware backtesting engine with point-in-time universe reconstruction and automated look-ahead/survivorship bias detection, stress-tested across an 18-fold robustness grid on 5 years of Nifty 50 data, with each experiment completing in under 3 minutes.

PRScope | *Python, PostgreSQL, FastAPI, GitHub OAuth, Typescript, React*

Live Demo | GitHub

- Built a Chrome extension and FastAPI-based platform that automatically reviews GitHub pull requests, combining AST analysis, dependency graphs, and call graph tracing to generate explainable risk scores.
- Built static security analysis pipelines to detect dependency risks, hardcoded secrets, SQL injection patterns, and other common vulnerabilities before code merges.
- Integrated GitHub OAuth, Jira requirement tracking, and a Gemini-powered assistant to auto-summarize diffs and generate confidence-ranked review comments.

BiasScope | *Python, Hugging Face Transformers, PostgreSQL, FastAPI, Next.js*

Live Demo | GitHub

- Built a distributed news intelligence platform that transforms multi-source media streams into structured claims, evidence graphs, and event-level intelligence.
- Designed a claim-centric data model linking claims, evidence, publishers, and events to enable cross-source corroboration and evidence-grounded analysis.
- Developed transformer-based NLP services for bias analysis, factual claim extraction, and source-reliability scoring, with asynchronous pipelines to process newly published content incrementally.

DataScope | *Python, Pandas, scikit-learn, PostgreSQL, FastAPI*

Live Demo | GitHub

- Built a machine learning governance platform that evaluates dataset quality, feature integrity, and deployment readiness through automated validation, leakage detection, and anomaly analysis.
- Built an explainability engine using SHAP attribution and permutation importance to trace model predictions back to root-cause features, flagging deployment risks before production rollout.
- Designed a consensus-based anomaly detection framework combining Isolation Forest, DBSCAN, and statistical methods (Z-score, MAD) to reduce false positives across heterogeneous datasets.
- Simplified dataset preparation by building one-click cleaning through a configurable 5-step pipeline, and added drift detection between dataset versions to catch data quality regressions before they reach production.

TECHNICAL SKILLS

Languages: C++, C, Python, SQL

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Machine Learning

Databases: MongoDB, PostgreSQL

ML & AI: NumPy, Pandas, Matplotlib, scikit-learn

Developer Tools & Frameworks: Git, GitHub, Linux, Nginx, Postman, Pytest, GitHub Actions

Backend & Infrastructure: FastAPI, REST APIs, Docker, Kubernetes, Redis, Celery, AWS Cloud (Fundamentals)

ACHIEVEMENTS

- Solved **250+** Data Structures & Algorithms problems in **C++** across LeetCode and other coding platforms.
- Achieved a peak **LeetCode Contest Rating of 1863 (Knight)** through consistent participation in weekly and biweekly coding contests.